Windows Server Exam Report

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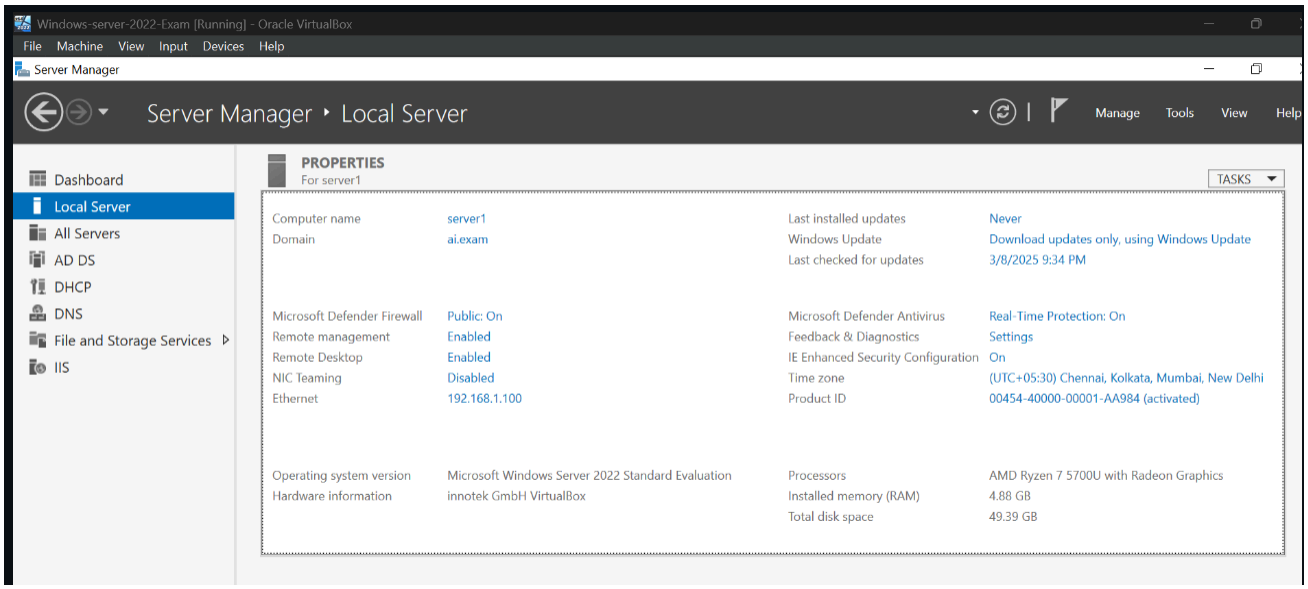
Question

Your organization is enhancing its Windows Server infrastructure by configuring a DHCP server for dynamic IP allocation, enabling internal DNS resolution, and hosting a secure website on IIS with enforced HTTPS connections and automatic HTTP-to-HTTPS redirection. As a Windows Server Administrator, set up the DHCP server to automatically update DNS records for DHCP clients within the "ai.exam" domain. Additionally, deploy the top 10 Group Policy Objects (GPOs) to strengthen security and optimize the operational efficiency of all systems in the "Sales" department.

Machine Details

Server

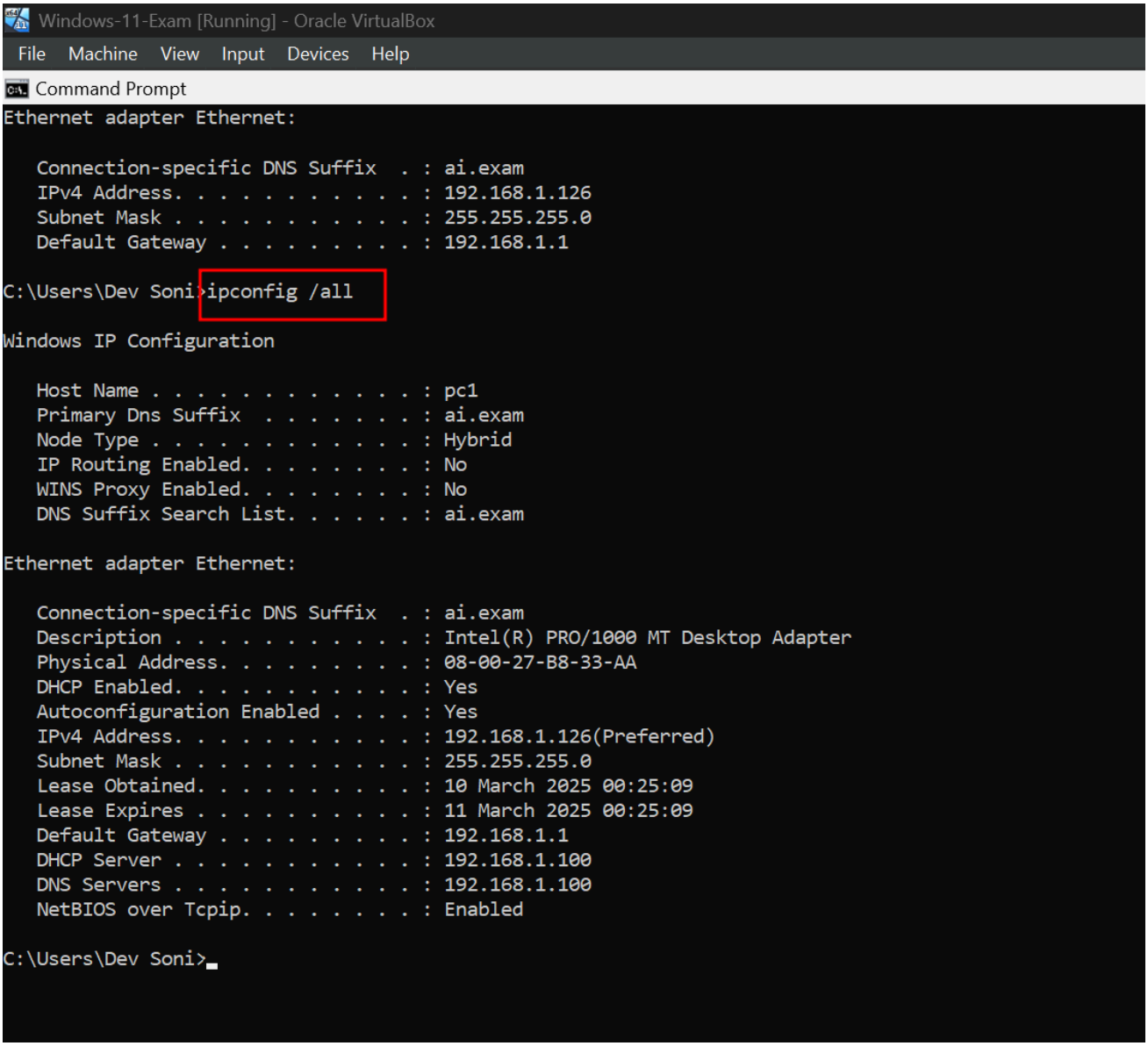
- **Operating System**: Windows Server 2022
- **Username**: Administrator
- **Password**: rish@bh123
- **Hostname**: server1
- **Domain**: ai.exam



Client 1

- **Operating System**: Windows 11
- **Local Admin**: Dev Soni
- **Password**: ex@m@Dev1

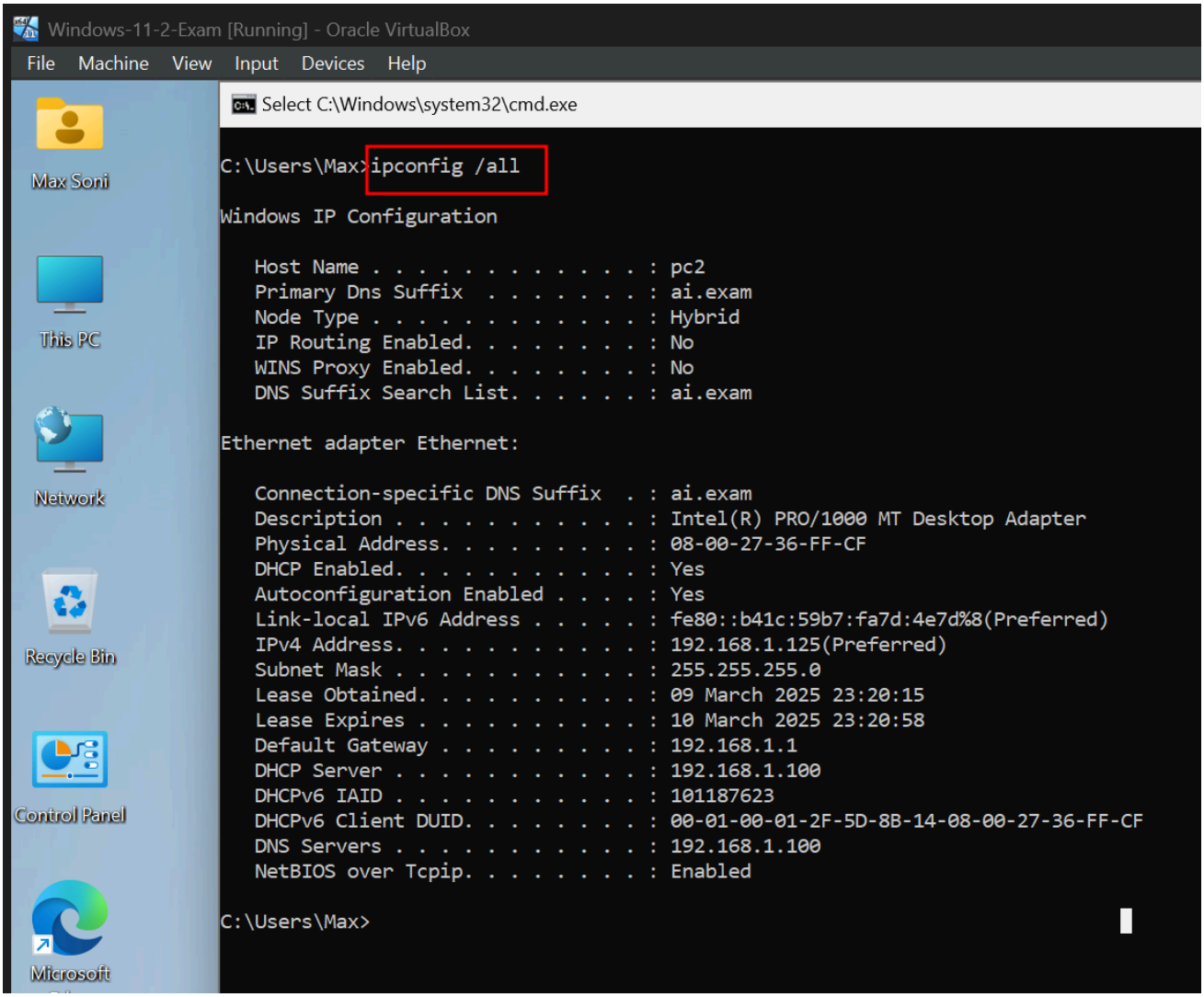
- **Hostname**: pc1



Client 2

- **Operating System**: Windows 11
- **Local Admin**: Max Soni
- **Password**: ex@m@Max1

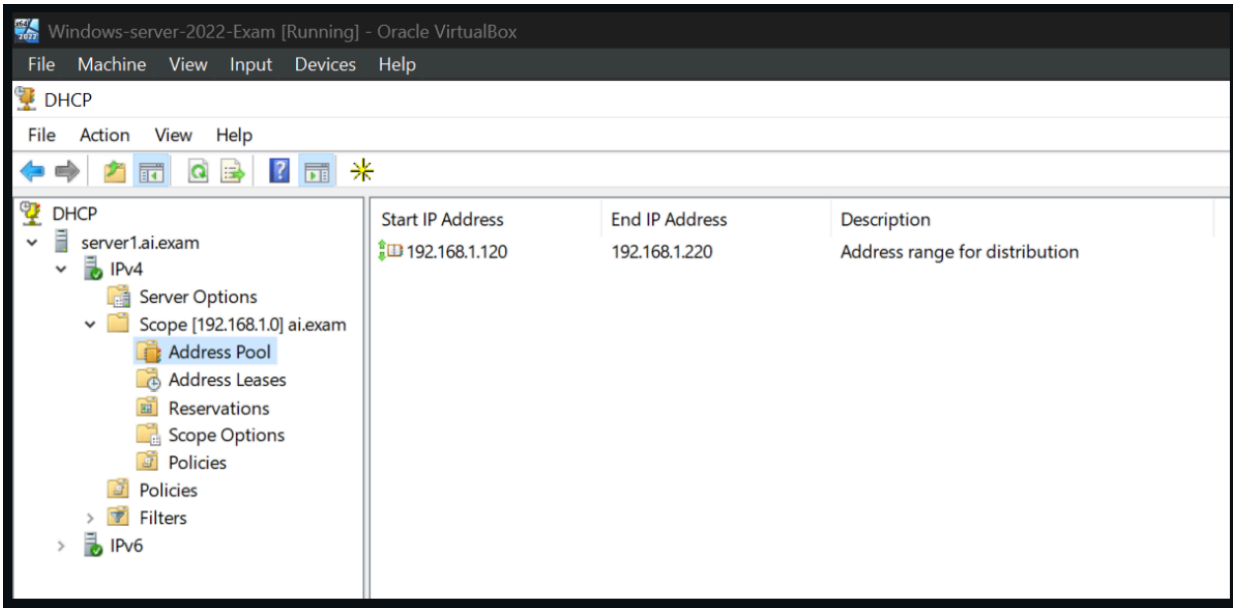
- **Hostname**: pc2



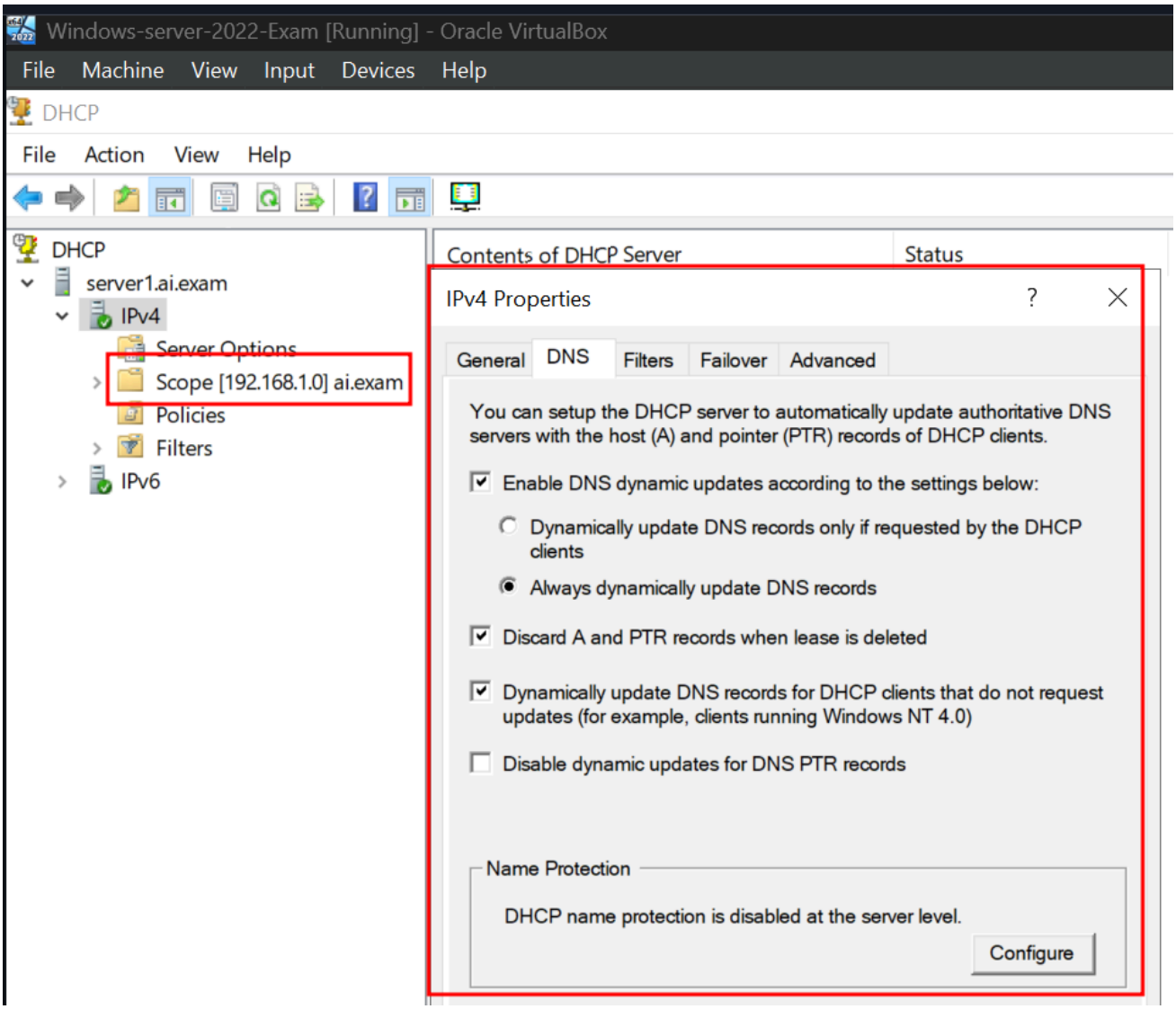
DHCP Configuration

- **Credentials to Authorize DHCP Server in AD DS**:
  - **User Name**: AI\Administrator
- **Scope Name**: ai.exam
- **IP Address Pool Range**:
  - Start: 192.168.1.120
  - End: 192.168.1.220
  - Subnet Mask: 255.255.255.0
- **Exclusion Range**: No exclusion
- **Lease Duration**: 1 day
- **DHCP Options**:
  - **Default Gateway**: 192.168.1.1

- **DNS Server**: 192.168.1.100



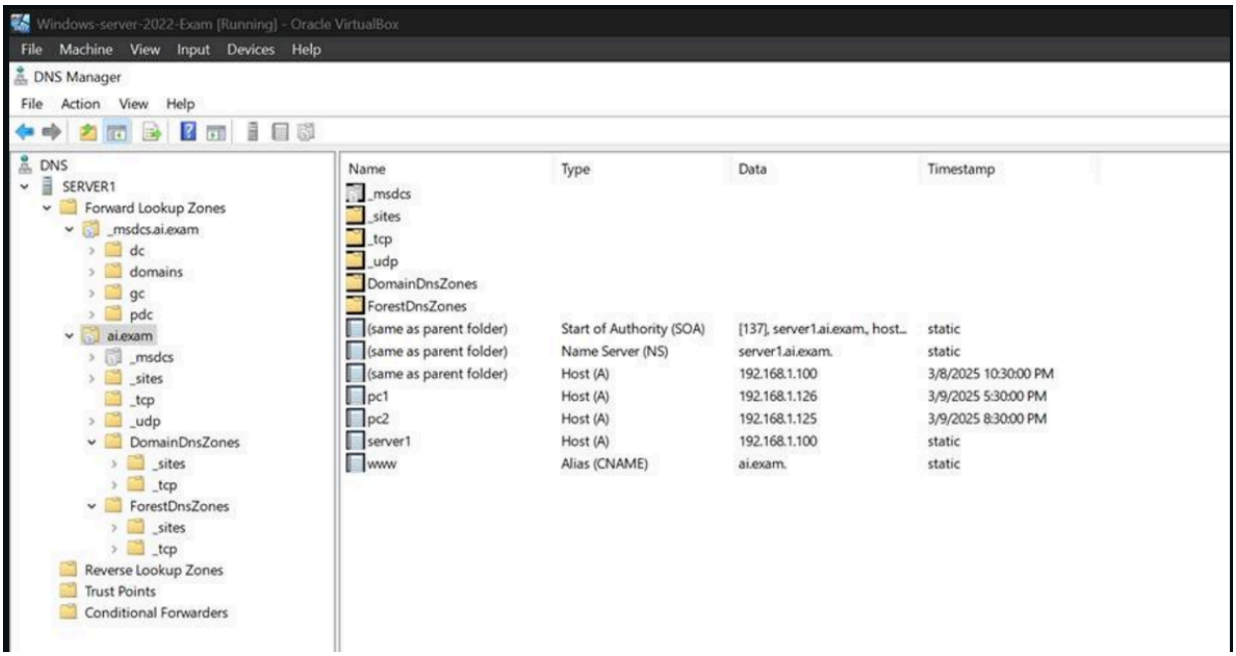
- **Automatic DNS Updates**: Configured to automatically update DNS records for DHCP clients.



DNS Configuration

- **Forward Lookup Zone**: ai.exam
- **Records**:
  - **Host (A) Records**:
    - ai.exam: 192.168.1.100
    - server1: 192.168.1.100
  - **CNAME Record**:

    - www → ai.exam



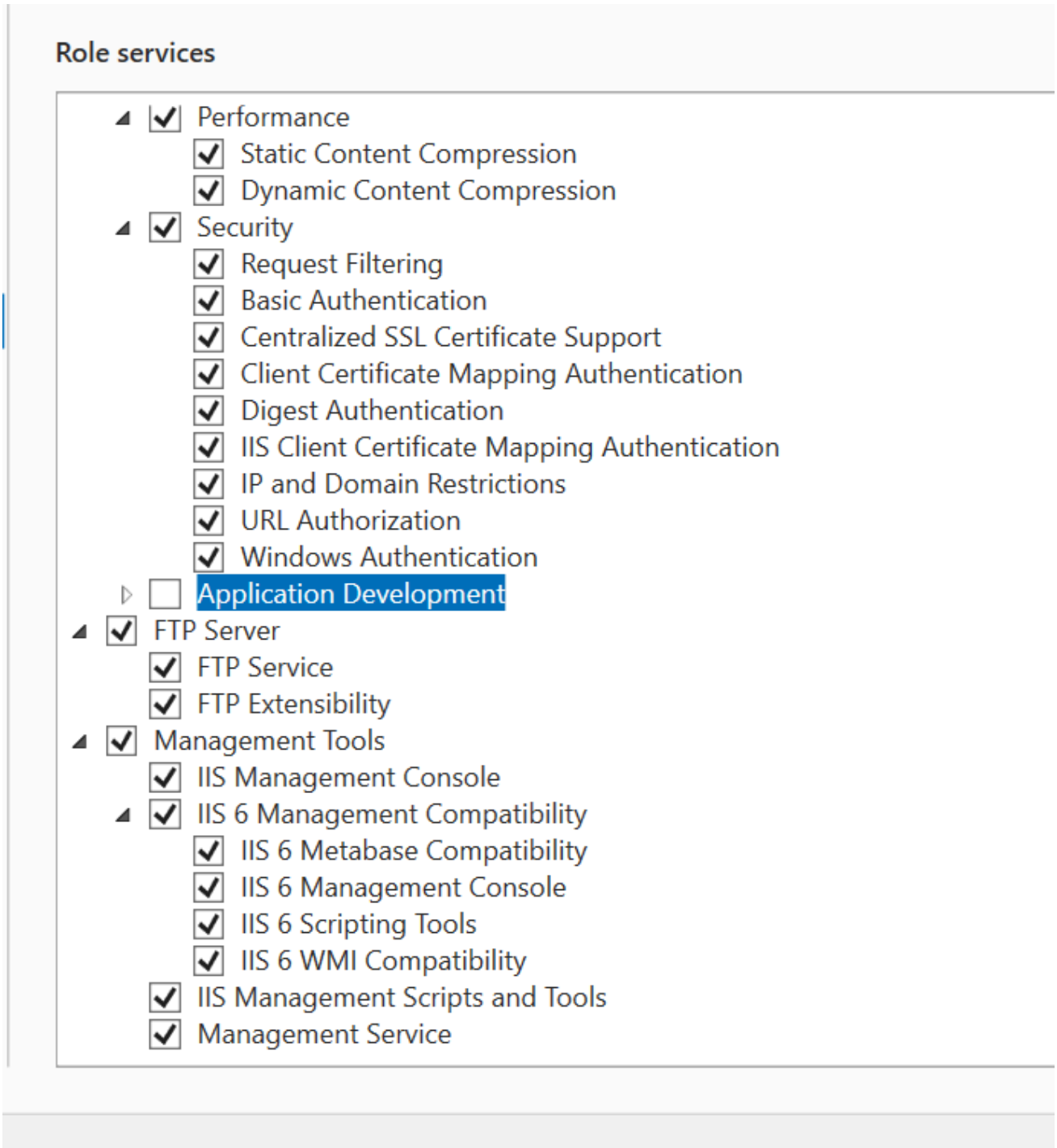
IIS Web Hosting

- **Roles Added**:

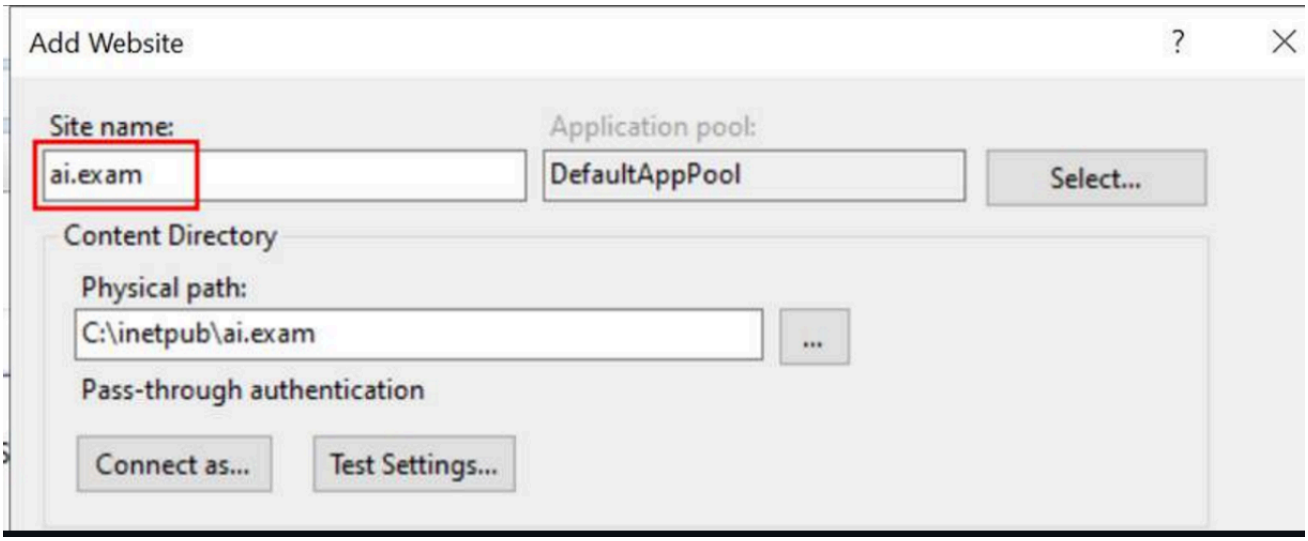
  - Web Server (IIS)

Role services

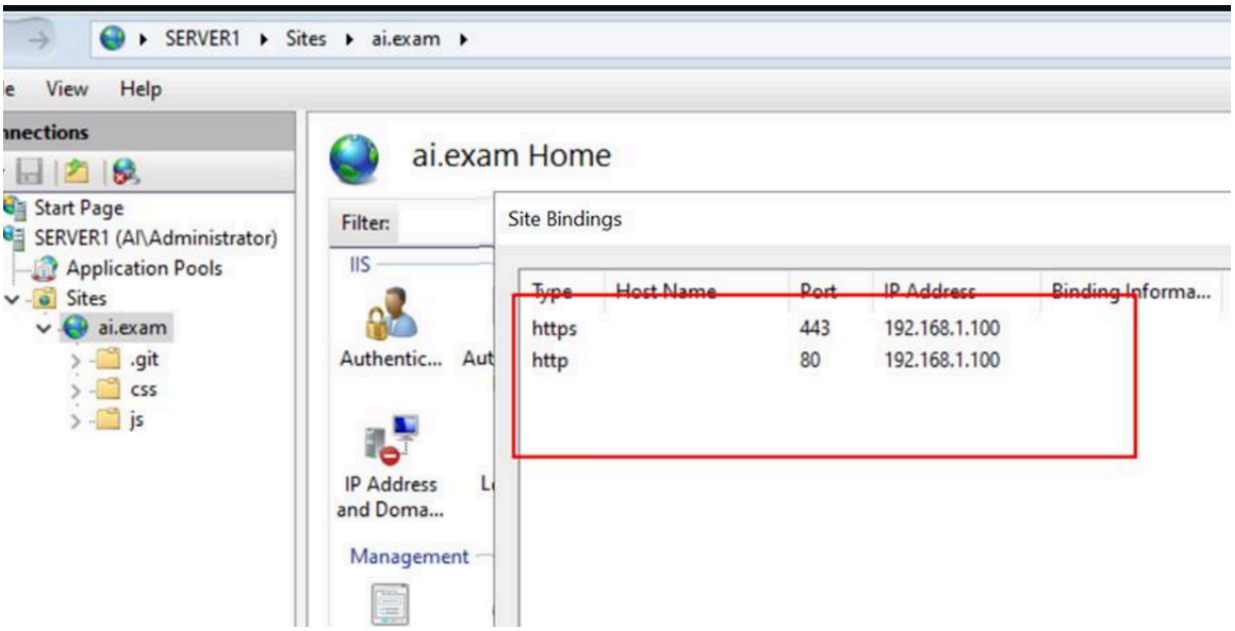
- [x] Web Server
  - [x] Common HTTP Features
    - [x] Default Document
    - [x] Directory Browsing
    - [x] HTTP Errors
    - [x] Static Content
    - [x] HTTP Redirection
    - [x] WebDAV Publishing
  - [x] Health and Diagnostics
    - [x] HTTP Logging
    - [x] Custom Logging
    - [x] Logging Tools
    - [x] ODBC Logging
    - [x] Request Monitor
    - [x] Tracing
  - [x] Performance
    - [x] Static Content Compression
    - [x] Dynamic Content Compression
  - [x] Security
    - [x] Request Filtering
    - [x] Basic Authentication
    - [x] Centralized SSL Certificate Support
    - [x] Client Certificate Mapping Authentication
    - [x] Digest Authentication
    - [x] IIS Client Certificate Mapping Authentication
    - [x] IP and Domain Restrictions



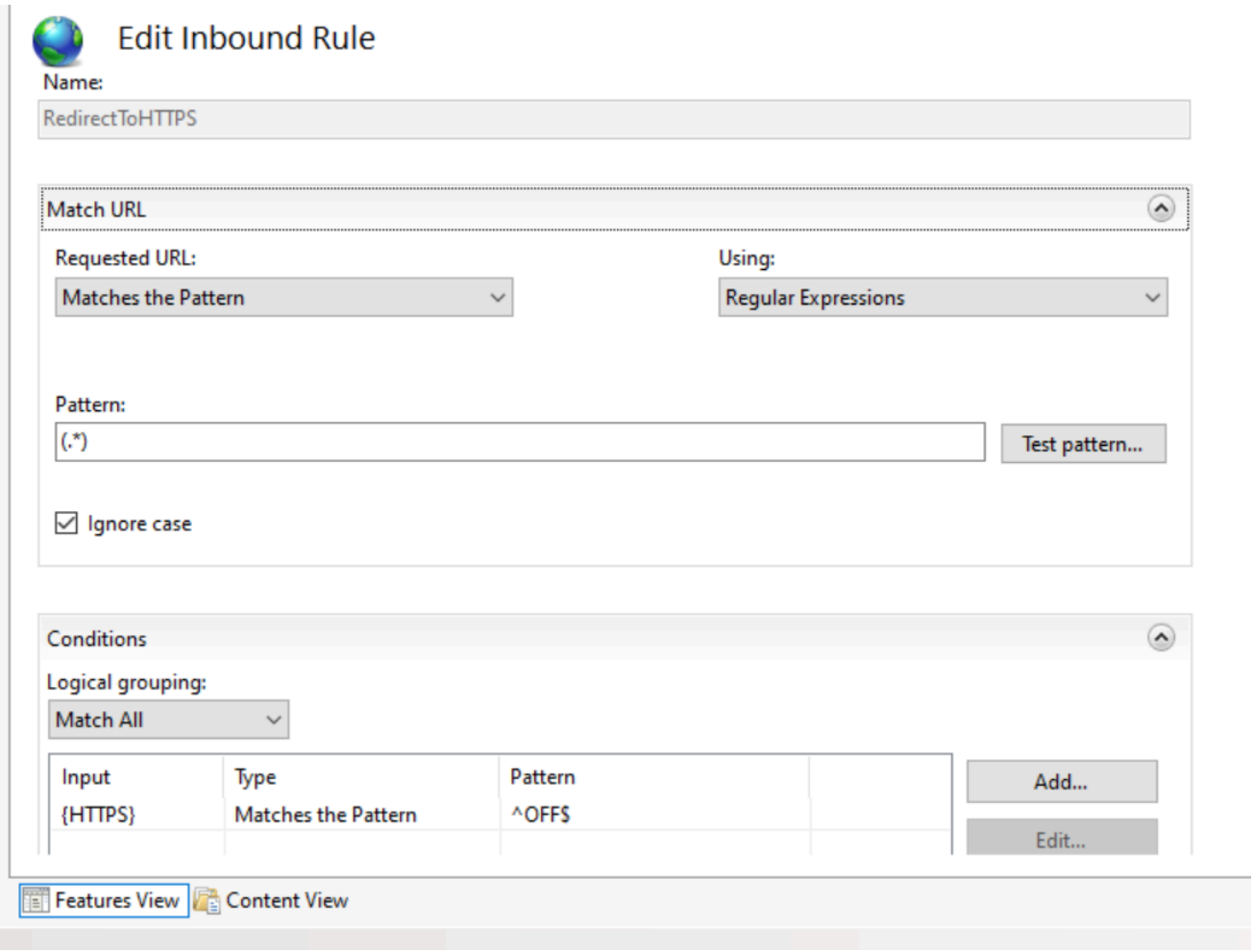
- **Website Name**: ai.exam
- **Physical Path**: C:\inetpub\ai.exam

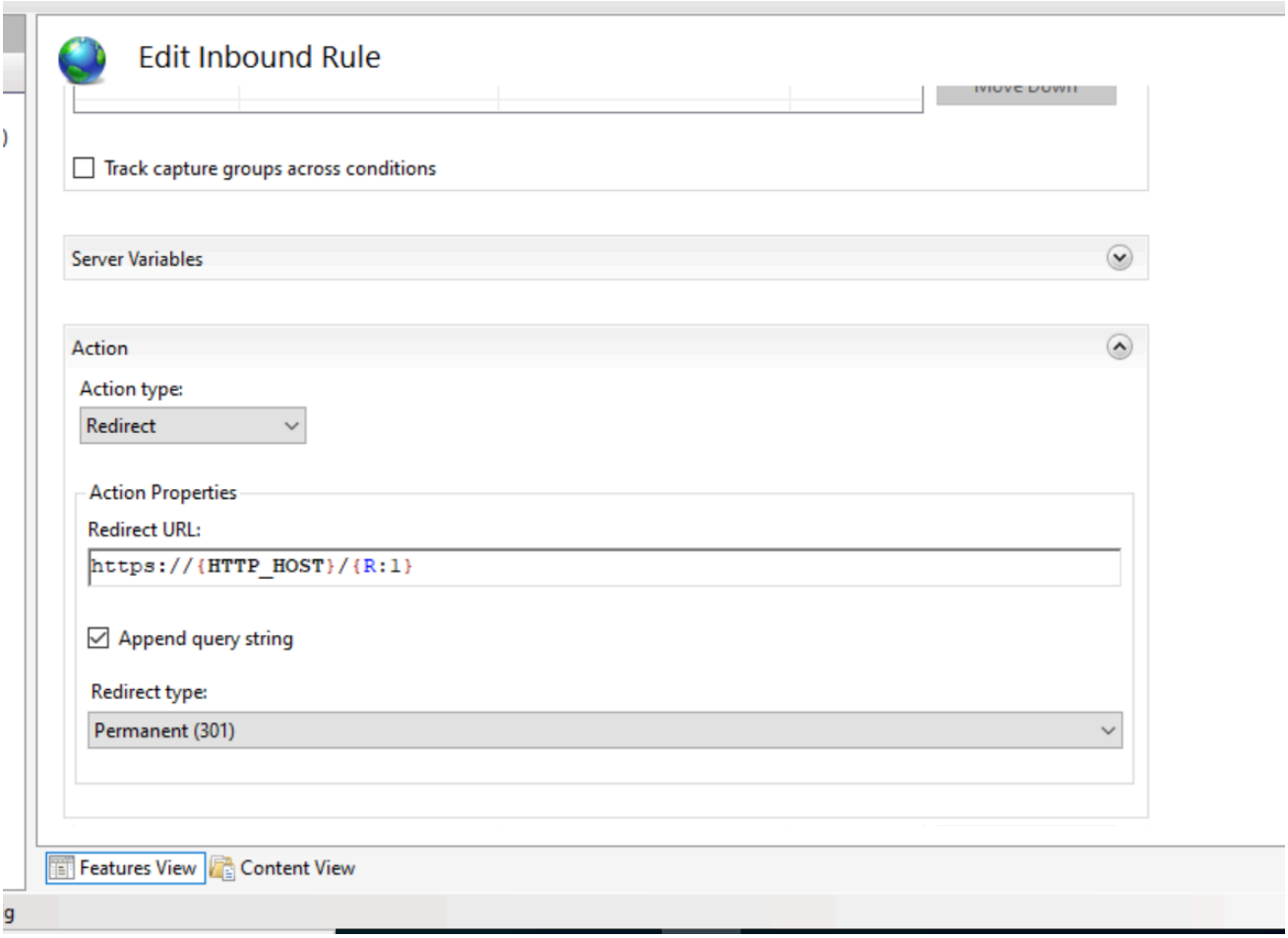


- **Bindings**:
  - HTTP: Port 80
  - HTTPS: Port 443



- **HTTP Redirection**: Redirects all HTTP requests to HTTPS.


Edit Inbound Rule
Name:
RedirectToHTTPS
Match URL
Requested URL:
Matches the Pattern
Using:
Regular Expressions
Pattern:
(.*)
Test pattern...
Ignore case
Conditions
Logical grouping:
Match All
Input
Type
Pattern
{HTTPS}
Matches the Pattern
^OFF$
Add...
Edit...
Features View
Content View



Top Group Policies on the Sales OU

1. **Disable Removable Storage (Block USB Drives)**
   - Path: Computer Configuration → Policies → Administrative Templates → System → Removable Storage Access
2. **Prevent Changing Desktop Background**
   - Path: User Configuration → Policies → Administrative Templates → Control Panel → Personalization
3. **Restrict Control Panel and PC Settings Access**
   - Path: User Configuration → Policies → Administrative Templates → Control Panel
4. **Restrict Software Installation**
   - Path: Computer Configuration → Policies → Windows Settings → Security Settings → Software Restriction Policies
5. **Password Policy (Enforce Strong Passwords)**
   - Path: Computer Configuration → Policies → Windows Settings → Security Settings → Account Policies → Password Policy
6. **Restrict Remote Desktop (RDP) Access**

- Path: Computer Configuration → Policies → Administrative Templates → Windows Components → Remote Desktop Services

7. **Block Remote Access to Registry Editor**
   - Path: Computer Configuration → Policies → Windows Settings → Security Settings → Local Policies → Security Options
8. **Restrict Access to Command Prompt (Prevent Running Scripts)**
   - Path: User Configuration → Policies → Administrative Templates → System
9. **Disable Guest Account**
   - Path: Computer Configuration → Policies → Windows Settings → Security Settings → Local Policies → Security Options
10. **Enable Windows Defender Firewall with Predefined Rules**
    - Path: Computer Configuration → Policies → Windows Settings → Security Settings → Windows Defender Firewall

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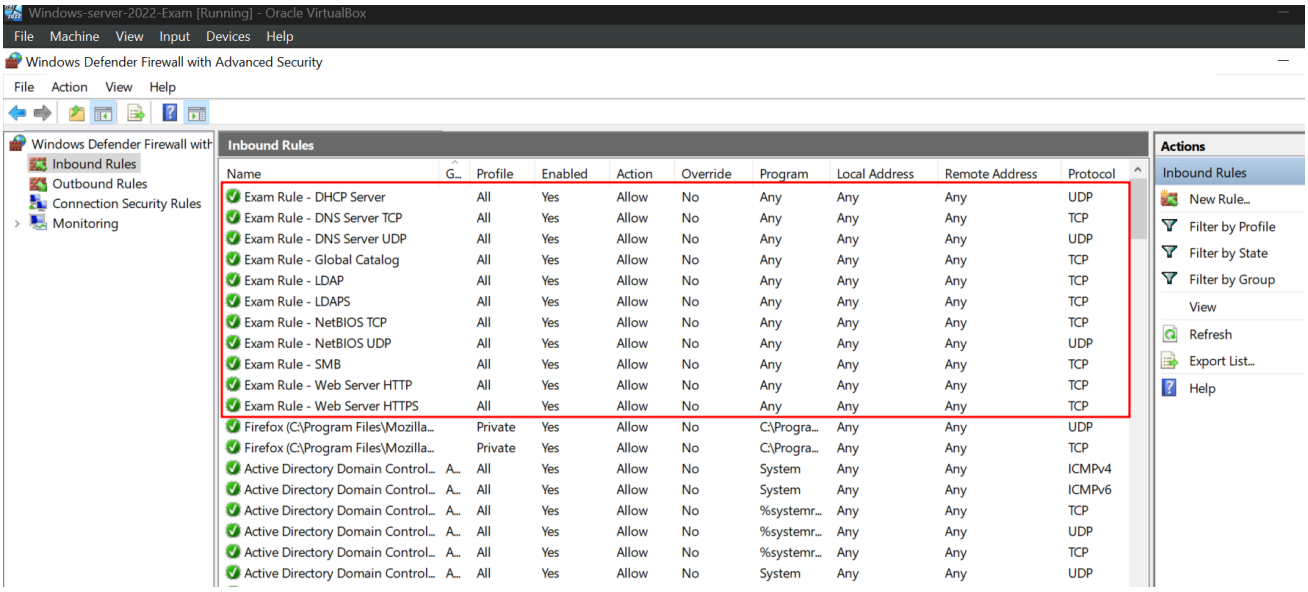
Firewall Configuration

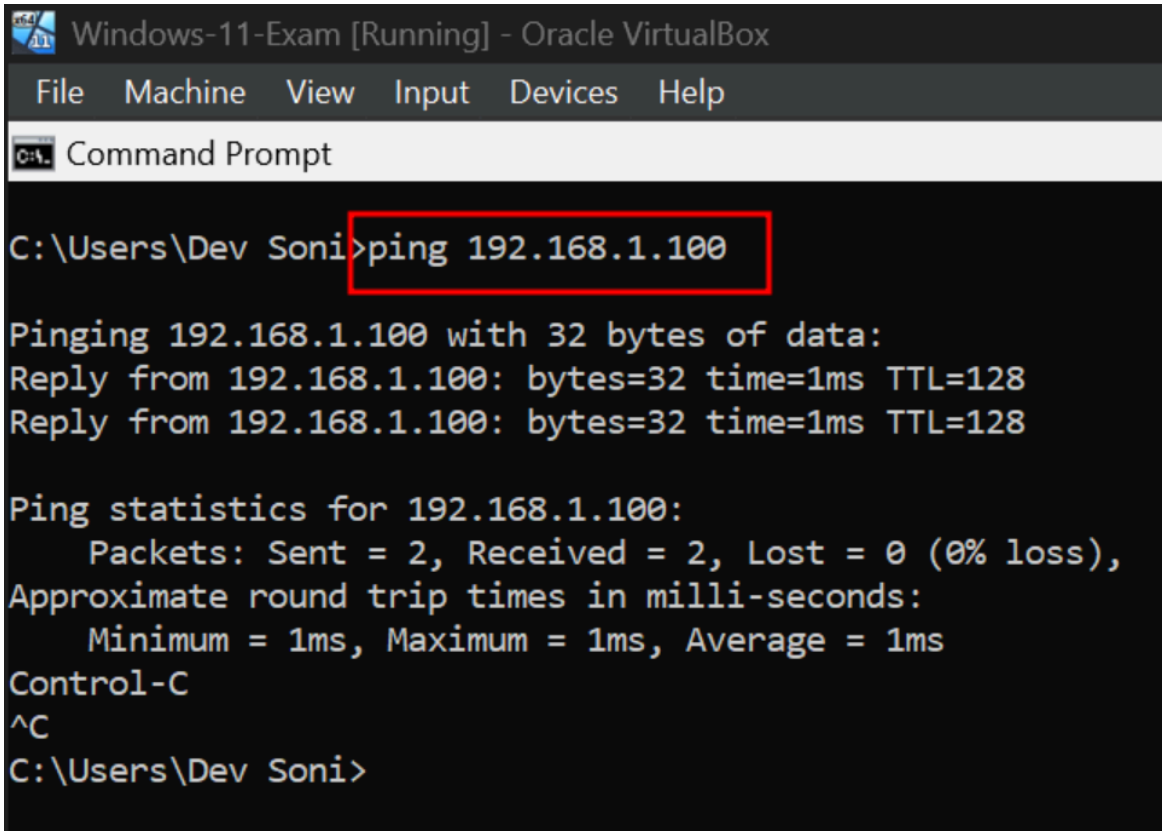
- **Inbound Rules**:



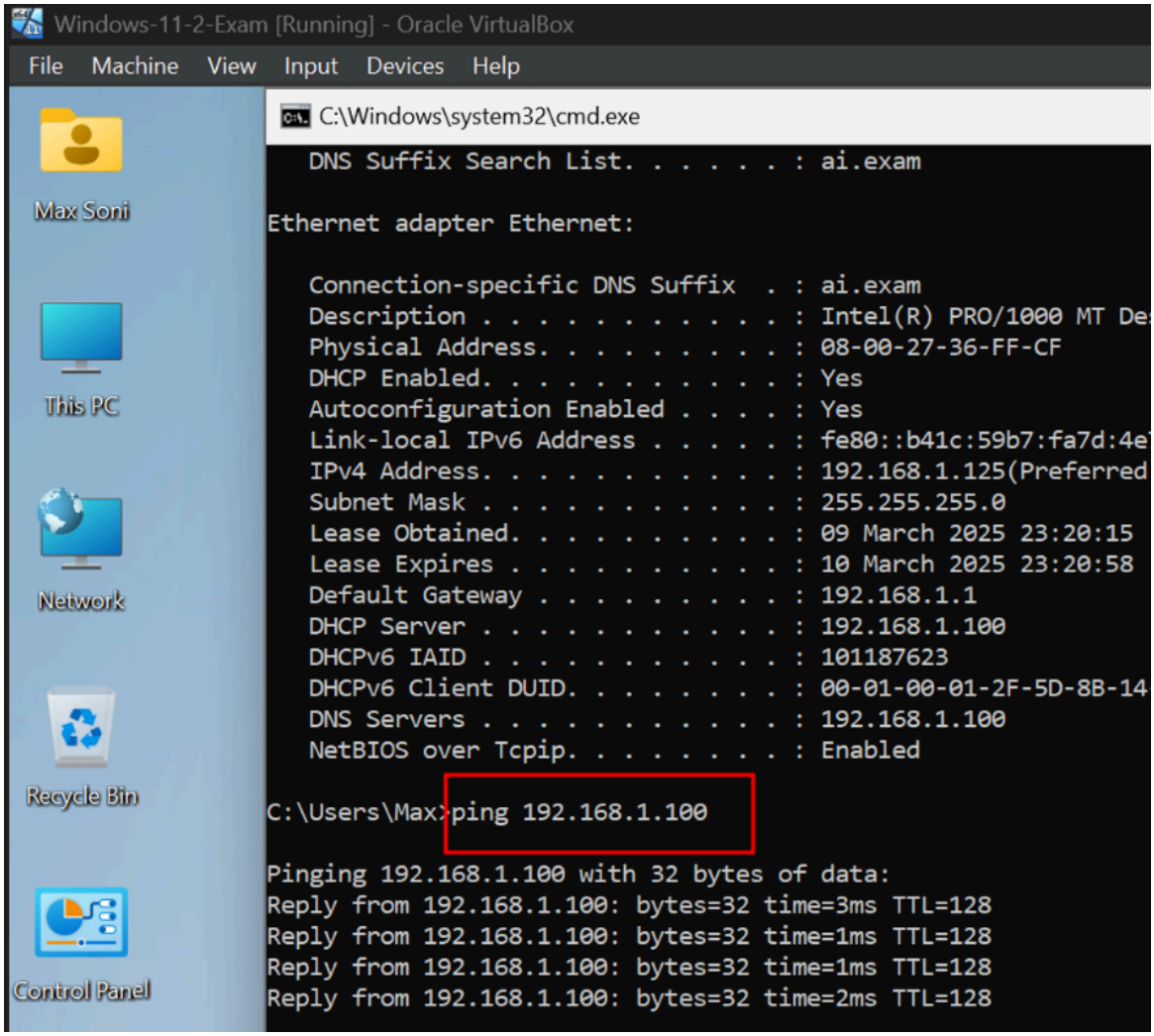
  - DHCP Server: Allowed
  - DNS Server TCP/UDP: Allowed
  - Web Server HTTP/HTTPS: Allowed
  - LDAP: Allowed
  - Global Catalog: Allowed

Connectivity Tests

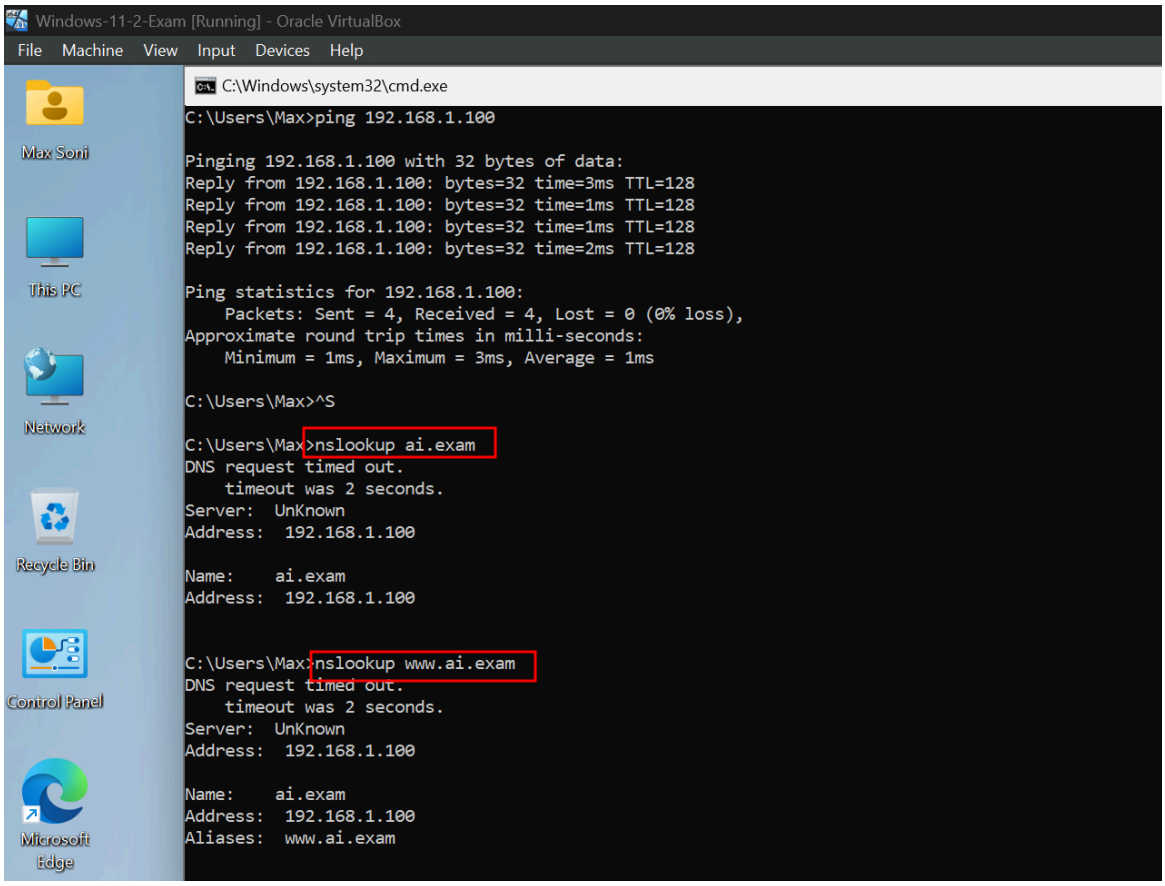
- **Ping Tests**:
  - Successful ping tests were conducted between clients and the server, confirming network connectivity.
    - **Client 1 (pc1)** to **Server**: Successful



- **Client 2 (pc2)** to **Server**: Successful

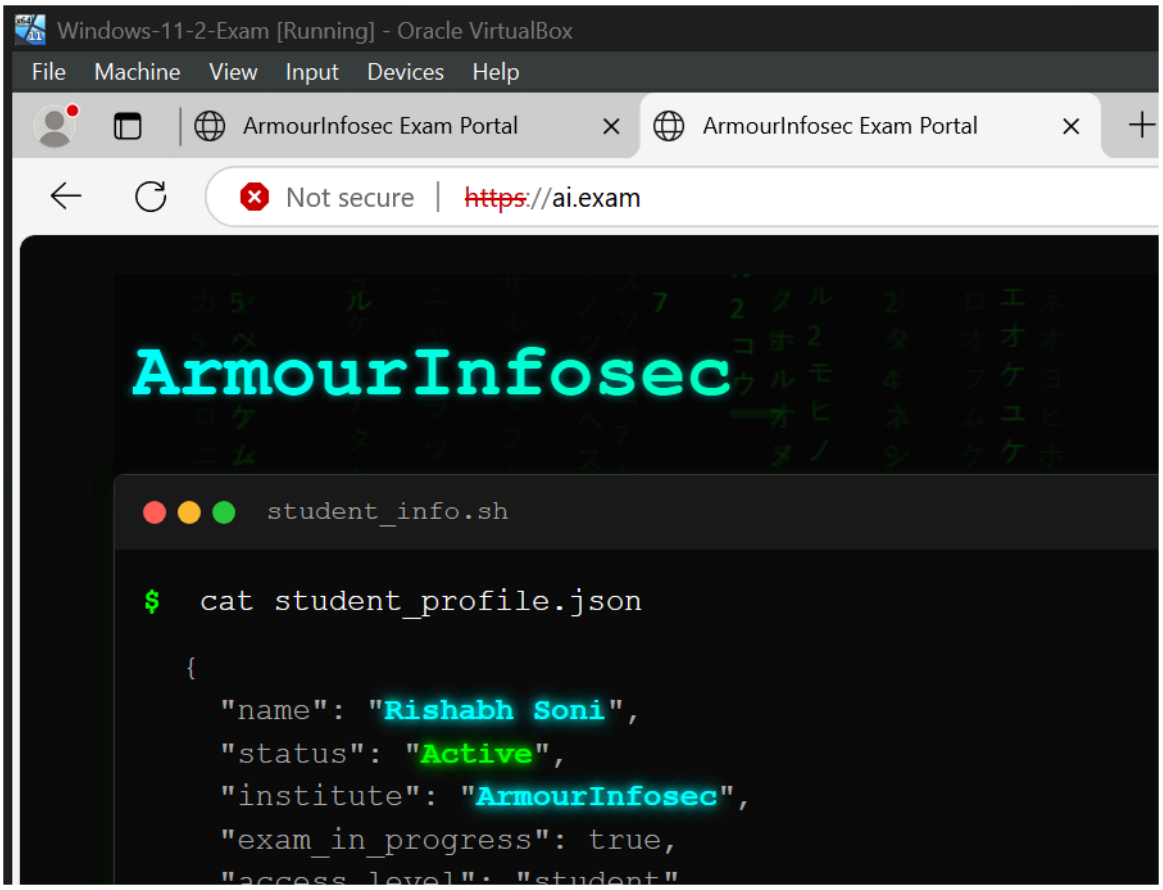


- **NSLookup**:
  - Successful DNS resolution for the following domains:
    - **ai.exam**: Resolved to `192.168.1.100`
    - www.ai.exam: Resolved to `192.168.1.100`



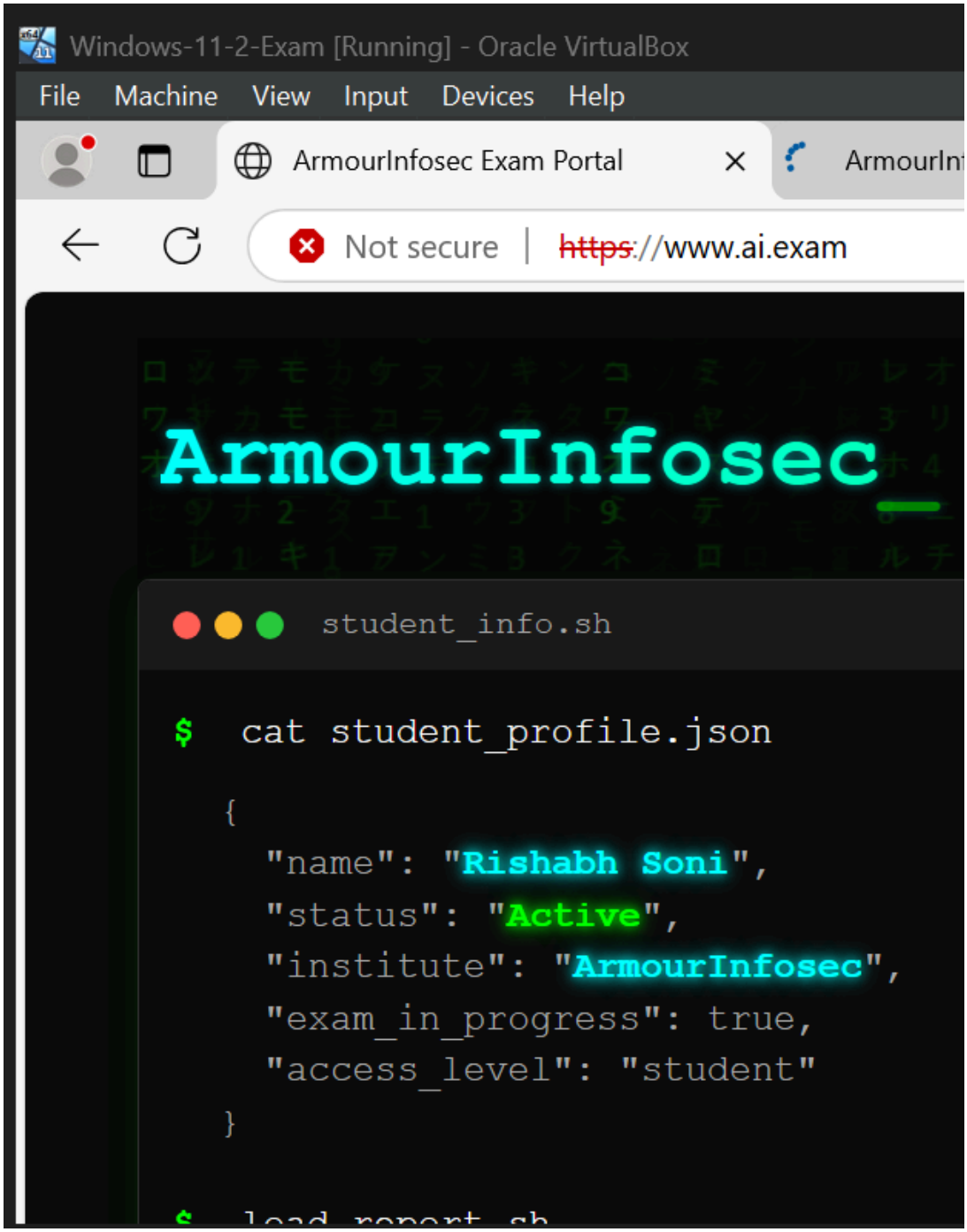
- **Web Access**:
  - Successful access to the website **ai.exam** via:
    - **HTTP**: Accessed at `http://ai.exam`

- **HTTPS**: Accessed at `https://ai.exam`



- **www.ai.exam via:
  - **HTTP**: Accessed at `http://www.ai.exam`

- **HTTPS**: Accessed at `https://www.ai.exam`



Local Account Passwords

Machine	Username	Password
Server 2022	ADMINISTRATOR	rish@bh123
pc1	Dev Soni	ex@m@Dev1
pc2	Max	ex@m@Max1

ADDS Usernames and Passwords

OU	Username	Password
IT	it1	rish@bh123
IT	it2	rish@bh123
Sales	sales1	rish@bh123
Sales	sales2	rish@bh123

Conclusion

- All configurations successfully implemented and tested.
- DHCP server is operational.
- DNS resolution is functioning correctly.
- IIS web hosting is active.
- Connectivity tests confirmed all systems are functioning as intended.
- Group Policy Objects deployed to enhance security and optimize efficiency in the "Sales" department.